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CYB 499: LLM and Cyber Security Applications

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February 12, 2026

Homework Assignment #1

Question 1: LLM Performance on Cybersecurity Multiple-Choice Questions

Setup

A set of 30 cybersecurity multiple-choice questions were selected from various CSSLP quizzes.

1. Question Set: questions.txt
2. Answer Key: answersforquestions.txt

Topics:

- Threat modeling (STRIDE, DREAD)
- Attack surface measurement
- Security reviews
- Secure Development Lifecycle
- Integrity, availability and non-repudiation

Models:

1. GPT-5-mini via OpenAI API (~149 billion parameters, instruction-tuned)
2. GPT-2 via HuggingFace Local (124 million parameters, pre-instruction tuned model)

Each model received the same question; responses were then recorded and compared against the answer key.

Results

Table 1

Model Performance on Cybersecurity Multiple-Choice Questions

Model	Parameters	Correct	Accuracy
GPT-5-mini	149 billion	27/30	90%
GPT-2	124 million	4/30	13.3%

Error Analysis

GPT-5-mini

The GPT-5-mini model only missed three questions, the missed questions had to do with governance and requirement classification, more specifically: security review purpose classification and requirement mapping distinctions. However, these incorrect answers do not seem to come from incorrectly understanding threat modeling frameworks or security principles. Instead, it seems to reflect subtle taxonomy level distinctions. Because the model correctly handled STRIDE definitions, DREAD scoring, attack surface measurement and more. Overall, it demonstrated strong domain comprehension and reliable structured reasoning.

GPT-2

The same cannot be said for GPT-2, which performed extremely poorly with the question set. The model seemed to default to answering "a)" and had many other problems including occasional truncated outputs, inconsistent formatting and little evidence of systematic reasoning. Its 13.3% accuracy indicates a patterned bias rather than informed selections. GPT-2 also struggled with multi-question stability and constrained multiple-choice formatting.

Model Interpretation

The difference in performance is primarily due to instruction tuning, alignment optimization and model scale. GPT-5-mini is a 149 billion parameter instruction-tuned model designed to follow structured prompts and operate within defined task constraints. It has been optimized not just to predict text but to respond in ways that satisfy explicit instructions, formatting requirements and structured evaluation scenarios. In contrast, GPT-2 is a 124 million parameter autoregressive transformer trained primarily for next-token prediction. It was in no way optimized for instruction following, constraint adherence or structured output reliability. Its objective during training was to continue text in a statistically plausible way, not to select correct answers within a constrained multiple-choice framework.

The scale difference also plays an important role. With a 149 billion parameter model it is going to have significantly greater representational capacity than a 124 million parameter model. Its huge parameter base allows it to encode more complex relationships between concepts such as STRIDE threat categories and SDL processes. Large parameter counts enable this deeper abstraction and improved contextual stability across multi-question tasks. However, scale alone does not explain the full gap in the comparisons. Instruction tuning and alignment mechanisms allow GPT-5-mini to interpret prompts as tasks rather than as open-ended text completion problems and this is the key difference. This distinction is critical also in cybersecurity evaluation settings where responses must conform to strict formatting and decision boundaries. The experiment suggests that instruction-aligned, large-scale models are substantially more reliable for structured cybersecurity evaluation tasks than the early-generation models. The observed 76.7% performance gap reflects both architectural maturity and alignment optimization, not simply incremental improvement.

Question 2: LLM generation of Multiple-Choice Questions

Setup

The document used for question generation was a basic rule book for Magic: The Gathering (36 pages long, MTGBASICS.PDF). Both models were instructed to generate 10 multiple-choice questions based only on the document's content.

1. Document: Magic: The Gathering Basics PDF (36 pages long)

Models:

1. GPT-4o-mini via OpenAI API (~8 billion parameters, instruction-tuned)
2. GPT-2 via HuggingFace Local (124 million parameters, pre-instruction tuned model)

Model Output Comparison

GPT-4o-mini:

- 10 properly formatted multiple-choice questions
- Four answer choices per question
- Clearly identified correct answers
- Questions aligned with the document's content
- Consistent structure throughout

The questions were usable and reflected comprehension of the source material.

Example Question Produced:

1. **What is the primary objective of Magic: The Gathering? **

A) Collecting cards

B) Reducing your opponent's life to 0

C) Building a sideboard

D) Winning tournaments

**Correct Answer: ** B) Reducing your opponent's life to 0

*Explanation: The main goal is to reduce your opponent's life from 20 to 0. *

GPT-2

- Inconsistent formatting
- Incomplete or malformed questions
- Topic drift
- Failure to provide four answer choices

It did not follow instructions to generate 10 complete multiple-choice questions.

Example Question Produced:

1)

What do I need? What does this mean when playing online or offline ? How many people will be able access my account if they have not already played before (or even after)? If there were no rulesets available that would allow me entry into tournaments , why am i getting banned so early?? Why should anyone care who has been invited by another player because their name appears as "A" instead than something else??? Is someone going through some

Temperature Sensitivity

Question 2 was repeated under the following temperature values of 0.0, 0.2, 0.8.

Temperature 0.0

1. GPT-4o-mini produced stable and deterministic outputs.
2. GPT-2 remained structurally inconsistent despite deterministic decoding

Temperature 0.2

1. GPT-4o-mini showed minor phrasing variation while maintaining structure.
2. GPT-2 outputs remained unstable.

Temperature 0.8

1. GPT-4o-mini demonstrated creativity but preserved formatting and coherence.
2. GPT-2's instability increased significantly.

Interpretation of Temperature Effects

Temperature influences randomness in token selection. However, the experiment shows that temperature does not fix the architectural limitations. Instruction-aligned models maintain structure even at higher temperatures, while non-aligned foundation models become even more unstable as randomness increases. So, in this case temperature modifies variability but it does not create reasoning capability where it does not exist.

Analysis

Several conclusions can be drawn from these results. First, GPT-5-mini achieved 90% accuracy on structured cybersecurity multiple-choice questions, indicating strong capability within constrained, certification-style evaluation tasks. This does not imply broad cybersecurity expertise, but it does demonstrate reliable performance under clearly defined decision boundaries. In contrast, GPT-2 showed consistent weakness in both answering structured questions and generating properly formatted multiple-choice content. Its failures were not isolated errors but systemic issues related to formatting instability, patterned bias and lack of task adherence. Across both experiments, instruction-tuned models consistently maintained structural integrity and followed explicit constraints, while GPT-2 frequently deviated from required formatting. Additionally, increasing temperature did not improve GPT-2's performance; rather the higher randomness amplified instability. The results suggest that modern instruction-tuned models are suitable for structured assessment tools, certification preparation (in this case CSSLP) and controlled educational content generation.

This assignment demonstrates a clear performance gap between modern instruction-aligned language models and early-generation foundation models in structured multiple-choice evaluation tasks. In Question 1, GPT-5-mini achieved 90% accuracy, while GPT-2 achieved 13.3%. In Question 2, GPT-4o-mini generated coherent, document-aligned multiple-choice questions, whereas GPT-2 failed to consistently adhere to formatting constraints. However, it is important to clarify scope. This experiment evaluated structured multiple-choice answering and question generation. It does not establish overall cybersecurity expertise, nor does it claim that modern models possess independent reasoning capability beyond constrained tasks. The results primarily demonstrate that instructional alignment and architectural maturity significantly improve reliability in structured evaluation settings. Early-generation foundation models were not designed for such constrained tasks and therefore performed poorly under these conditions. Overall, instruction-tuned models show strong potential for structured educational and assessment applications, while legacy foundation models are not suitable for constraint-based evaluation environments.